Logger Library

Problem Statement

Requirements

- **You have to design and implement a logger library that helps applications log messages.**
- Client/application make use of your logger library to log messages to a sink
- Message
    - has **content** which is of type string
    - has a **level** associated with it
    - is directed to a destination (**sink**)
    - has **namespace** associated with it to identify the part of application that sent the message
- Sink
    - This is the destination for a message (eg text file, database, console, etc)
    - Sink is tied to one or more message level
    - one or more message level can have the same sink
- Logger library
    - Accepts messages from client(s)
    - Routes messages to appropriate sink
    - Supports following message level: DEBUG, INFO, WARN, ERROR, FATAL
    - enriches message with current timestamp while directing message to a sink
    - requires configuration during setup
- Sending messages
    - Sink need **not** be mentioned while sending a message to the logger library.
    - A message level is tied to a sink.
    - You specify message content, level and namespace while sending a message
- Logger configuration (see sample below)
    - Specifies all the details required to use the logger library.
    - Example:
        - time format
        - logging level
        - sink type
        - details required for sink (eg file location))
            - this depends on sink type

Sample Config1: text file as sink

```
ts_format:ddmmyyyyhhmmss
log_level:INFO sink_type:FILE
file_location:/var/log/app/info.log
```

```
(optional)thread_model:SINGLE
(optional)write_mode:SYNC
```

Sample Config2: database as sink

```
ts_format:dd:mm:yyyy hh:mm:ss
log_level:ERROR sink_type:DB
dbhost:<ip address>
dbport:<db port>
(optional)thread_model:MULTI
(optional)write_mode:ASYNC
```

Sample execution

```
INFO  [2022-06-27 11:14:44,942] Empty txnIds Nothing to fetch
WARN  [2022-06-27 11:28:06,229] No user found for the phone number
```

Log Rotation

```
application.log  application.log.2.gz  application.log.1.gz
```

- **Expectations / Goals**
  - Design your logger library to allow users to provide their own implementation of sink
  - You must provide implementation of the text file sink.
  - Add auto-rotate for log files based on file size. Compress previous log files.
- **Guidelines**
  - You may consider logger configuration as a key-value pair
  - Please use structures provided by the language of your choice.
  - Usage of any logging library as a part of implementation is prohibited
- **How you will be evaluated**
  - Separation of concerns
  - Abstractions
  - Application of OO design principles
  - Testability
  - Code readability
  - Language proficiency
- **Good to Have / Extensions, if any**
  - Addition of trackingId, hostName/instanceId in logs
  - Implementation of any other sink
- **How you will be evaluated**
  - Separation of concerns

- Abstractions
- Application of OO design principles
- Testability
- Code readability
- Language proficiency